Name (Print): ____________________________________

This exam contains 11 pages (including this cover page) and 9 problems. Check to see if any pages are missing. Print your name above.

You are allowed one sheet of notes. Turn in your notesheet along with your response. You are also allowed to use a calculator on the exam.

Read the following instructions carefully:

- You can only use methods discussed in class so far. i.e., you may **NOT** use any method we haven't learned in class so far.
- **Show ALL your work!** You are being assessed on your work and explanations. Correct answers with little to no work or explanation will not receive full credit. I give partial credits so don't leave a blank if you have some ideas.
- **Organize your work.** Your work should be clearly written and organized. Please use correct mathematical notations when appropriate. For example, please use vector notations when working with vectors and don't use vector notations when working with scalars and points.
- By submitting this exam, you agree that all work represents your individual effort and understanding. Failure to comply with any of the above statements will be considered a violation of the Student Code of Conduct and will result in a 0 for this assessment and possible disciplinary action.

Problem	Points	Score
1	14	
2	12	
3	12	
4	12	
5	12	
6	12	
7	12	
8	14	
9	0	
Total:	100	

Do not write in the table to the right. Adding the bonus, the total score won't exceed 100%.

Please keep in mind that an assessment is just another means of communication between you and me to help me understand what you have learned so far. It is not a measure of your worth as a person nor your intelligence or aptitude as a student. Just give this your best try.

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1. (14 points) Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be a linear transformation defined by $T(\mathbf{x}) = A\mathbf{x}$, where

$$A = \begin{bmatrix} 1 & -2 & 3 & 1 \\ 2 & -3 & 4 & 3 \\ -1 & 4 & -7 & 1 \end{bmatrix}$$

(a) Find the value of $T\left(\begin{bmatrix} 2 \\ -1 \\ 0 \\ 4 \end{bmatrix}\right)$.

(b) Is $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$ an image of this linear transformation? Justify your reasoning.

2. (12 points) Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear transformation such that

$$T\left(\begin{bmatrix} 4 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} \qquad \text{and} \qquad T\left(\begin{bmatrix} 5 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}$$

Find the standard matrix A that defines this transformation. That is, find the matrix A such that $T(\mathbf{x}) = A\mathbf{x}$.

3. (12 points) Let A, B, and C be $n \times n$ matrices such that

$$\det A = 4\,, \qquad \det B = -2\,, \qquad \text{and} \qquad \det C = 5$$

Evaluate the value of $\det\left(C^2 A B^{-1} A^T C^{-1}\right)$

4. (12 points) Let $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 2 & 2 & 4 \\ 0 & -1 & 3 & 2 \\ 2 & 4 & -2 & 1 \end{bmatrix}$ be a 4×4 matrix.

(a) Compute $\det A$.

(b) Is A an invertible matrix based on your answer in part (a)? Justify your reason.

5. (12 points) Find an equation of the plane passing through the points

$$(1,2,1)\,,\qquad (3,1,1)\,,\qquad \text{and}\qquad (0,3,2)$$

6. (12 points) Determine whether the lines

$$\mathbf{p}_1 = \begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} \qquad \text{and} \qquad \mathbf{p}_2 = \begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix} + s \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$$

are parallel, intersecting, or skew. If they intersect, determine the point of intersection.

7. (12 points) Is the following set of vectors linearly independent?

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ -3 \\ 4 \\ -5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ -8 \\ 9 \\ -13 \end{bmatrix} \right\}$$

If they are not linearly independent, find the *smallest* subset that spans the same subspace.

8. (14 points) Let $A = \begin{bmatrix} 1 & -2 & -4 & -1 & 3 \\ 2 & -3 & 5 & 0 & 2 \\ -3 & 10 & -24 & 11 & -25 \end{bmatrix}$.

(a) Find a basis for Col$A$. Then determine its dimension.

(b) Find a basis for Nul$A$. Then determine its dimension.

9. (10 points (bonus)) **This is a BONUS problem.**

Determine whether the following statements are true. Justify your reasoning.

(a) $\mathbb{R}^2 = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \mid x, y \in \mathbb{R} \right\}$ is a subspace of $\mathbb{R}^3$.

(b) $\left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} \mid x, y \in \mathbb{R} \right\}$ is a subspace of $\mathbb{R}^3$.